Software Requirements Specification (SRS)

Collab Sphere – Faculty Student Project

Prepared by: Team 46

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1. Introduction

CollabSphere is a web-based platform designed to facilitate structured collaboration between faculty members and students within academic institutions. The platform aims to modernize the process of project allocation by replacing manual coordination, email communication, and informal discussions with a centralized digital system.

In traditional academic environments, project opportunities are often shared through word of mouth or traditional communication channels. This makes it difficult for students to discover relevant projects and faculty members to evaluate potential applicants effectively. CollabSphere solves this problem by creating a transparent platform where faculty members can publish project opportunities, and students can browse, apply, and track their applications.

The system enables better collaboration, improves transparency, and ensures that academic projects are managed in an organized and structured manner. This document describes the functional and non-functional requirements for the CollabSphere platform and serves as a reference for developers, testers, instructors, and project stakeholders.

1.1 Document Purpose

The purpose of this Software Requirements Specification (SRS) document is to describe the functional and non-functional requirements of the CollabSphere platform. This document provides a clear and detailed description of system behavior, features, and constraints to guide the development, testing, and deployment of the system.

The SRS serves as a reference for developers, project stakeholders, instructors, and testers to ensure that the system meets its intended objectives and functions as required.

1.2 Product Scope

CollabSphere is designed to streamline the academic project allocation process within educational institutions. The system allows faculty members to publish project opportunities with detailed descriptions, required skills, eligibility criteria, and deadlines. Students can browse these opportunities, evaluate project requirements, and apply to projects that align with their interests and skills.

The platform also provides tools for faculty members to review student applications, evaluate candidates, and track project progress. Students maintain profiles that include academic records, skills, achievements, and links to professional platforms such as GitHub and LinkedIn.

The main objectives of the platform include:

- Providing a centralized platform for project collaboration
- Improving transparency in the project allocation process
- Simplifying communication between students and faculty
- Maintaining structured records of projects and evaluations
- Supporting efficient project discovery through categorization and filtering

By implementing these features, CollabSphere enhances academic collaboration and improves the overall experience of project management within institutions.

1.3 Intended Audience and Document Overview

- Developers – to understand system requirements and implement functionality.

- Project Team Members – to guide development and maintain consistency.
- Instructors and Evaluators – to assess project design and implementation.
- Testers and QA Engineers – to verify system functionality against requirements.
- The document is organized as follows:
- Introduction – Overview of the project.
- Overall Description – High level system description.
- Specific Requirements – Functional and interface requirements.

1. Non-functional Requirements – *Performance, security, and quality attributes.*
2. Additional Requirements – *Supporting information.*

2) Overall Description

2.1 Product Overview

The system enables faculty members to publish project opportunities with detailed descriptions and eligibility criteria. Students can browse these projects and submit applications. Faculty members can review applications and select suitable candidates for their projects.

The system follows a **layered architecture model**, consisting of the following layers:

1. **User Layer**
   This layer includes the system users: students, faculty members, and administrators.
2. **Presentation Layer**
   This layer provides the web interface through which users interact with the system.
3. **Application Layer**
   This layer contains backend services responsible for processing requests and implementing business logic.
4. **Service Layer**
   This layer includes system modules such as project management, application processing, and evaluation systems.
5. **Data Layer**
   This layer manages data storage using databases and document storage systems.

This architecture ensures scalability, maintainability, and modular system design.

2.2 Product Functionality

- Student registration and profile management
- Faculty project posting and management
- Project browsing and filtering
- Student project applications

- Faculty application review and evaluation
- Internal verification and discrepancy flagging
- Project categorization and recommendations
- Communication between students and faculty
- Project archiving and report generation
- Role-based access control

2.3 Design and Implementation Constraints

- Must be implemented as a web-based application
- Must support common web browsers
- Must implement role-based access control
- Database must securely store academic records
- Development must follow UML modeling standards
- Development timeline limited to 8 weeks
- Security and privacy of academic data must be ensured

2.4 Assumptions and Dependencies

- Faculty and students will actively use the system.
- Institutional servers and infrastructure will support deployment.
- Users will provide accurate academic information.
- Developers have sufficient expertise in web development technologies.
- The institution will support testing and evaluation activities.

3) Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

Login page

Student dashboard

Faculty dashboard

Project listing page

Application submission page

Profile management page

3.1.2 Hardware Interfaces

Desktop computers

Laptops

Web servers

Database servers

3.1.3 Software Interfaces

Web browsers

Backend server framework

Database management system

Authentication services

3.2 Functional Requirements

F1 – The system shall allow students and faculty to securely log in.

F2 – The system shall allow students to create and maintain profiles.

F3 – Faculty shall be able to create and manage project postings.

F4 – Students shall be able to browse available projects.

F5 – Students shall be able to apply for projects with supporting documents.

F6 – Faculty shall be able to review and evaluate applications.

F7 – Faculty shall be able to provide structured evaluation feedback.

F8 – Faculty shall be able to flag discrepancies in student information.

F9 – The system shall allow communication between students and faculty.

F10 – Completed projects shall be stored for future reference.

F11 – The system shall enforce role-based access permissions.

4) Other Non-functional Requirements

4.1 Performance Requirements

P1 – The system shall support **at least 500 concurrent users** accessing the platform simultaneously without significant degradation in performance.

P2 – The system shall return **project search results within 2 seconds** under normal network conditions.

P3 – The system shall process **student project applications within 3 seconds** after submission.

P4 – The system shall load dashboard pages within **2–4 seconds**.

P5 – The system shall handle **large numbers of project postings** without affecting response time.

P6 – The system shall maintain **database query response times under 1 second** for standard operations.

P7 – File uploads such as resumes and transcripts should be processed within **5 seconds** for files up to 5MB.

P8 – System downtime should not exceed **1% of total operational time** during an academic semester.

4.2 Safety and Security Requirements

- Secure login authentication
- Role-based access control
- Encrypted storage of sensitive data
- Prevention of unauthorized access
- Secure handling of uploaded documents

4.3 Software Quality Attributes

Reliability

The system shall maintain stable operation with minimal downtime and ensure accurate storage of user data.

Maintainability

The system shall follow modular architecture to allow easy updates and feature enhancements.

Usability

The platform will provide an intuitive user interface allowing students and faculty to navigate easily without extensive training.

5) Other Requirements

- *Must store student profiles*
- *Must store project details*
- *Must store application records*
- *Must store evaluation feedback*
- *Academic data privacy must be maintained*

Appendix A – Data Dictionary

Data Item	Description
StudentID	Unique identifier for students
FacultyID	Unique identifier for faculty
ProjectID	Unique project identifier
ApplicationID	Unique application record
EvaluationScore	Faculty evaluation value

Appendix B – Group Log

- System architecture discussions
- Module implementation planning
- Interface design meetings
- Testing strategy planning